

Hybrid FL: Algorithms and Implementation

Xinwei Zhang* Wotao Yin† Mingyi Hong* Tianyi Chen#

*University of Minnesota

† UCLA

RPI

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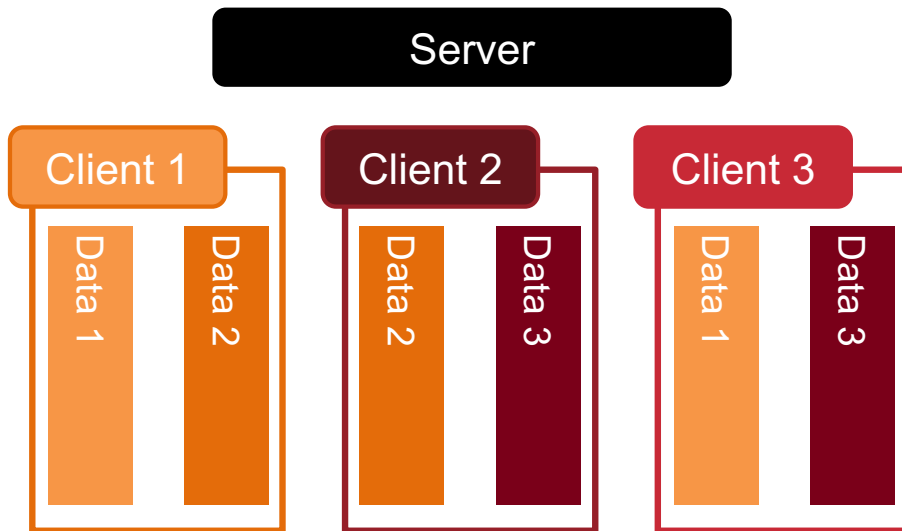


Introduction

- Horizontal FL & Vertical FL
- Limitations of HFL & VFL
- Hybrid FL - Setting & Challenge



❑ Horizontal & Vertical FL



Horizontal FL

- Each client: full feature, full model, **partial** data id
- Application
 - Google Keyboard [1]
 - Image classification [2]
- Algorithm
 - FedAvg [3], Scaffold [4], ...

[1] Bonawitz K, Eichner H, Grieskamp W, et al. Towards federated learning at scale: System design. arXiv 2019.

[2] Li W, Milletari F, Xu D, et al. Privacy-preserving federated brain tumour segmentation, MLMI 2019 .

[3] Li X, Huang K, Yang W, et al. On the Convergence of FedAvg on Non-IID Data, ICML 2019.

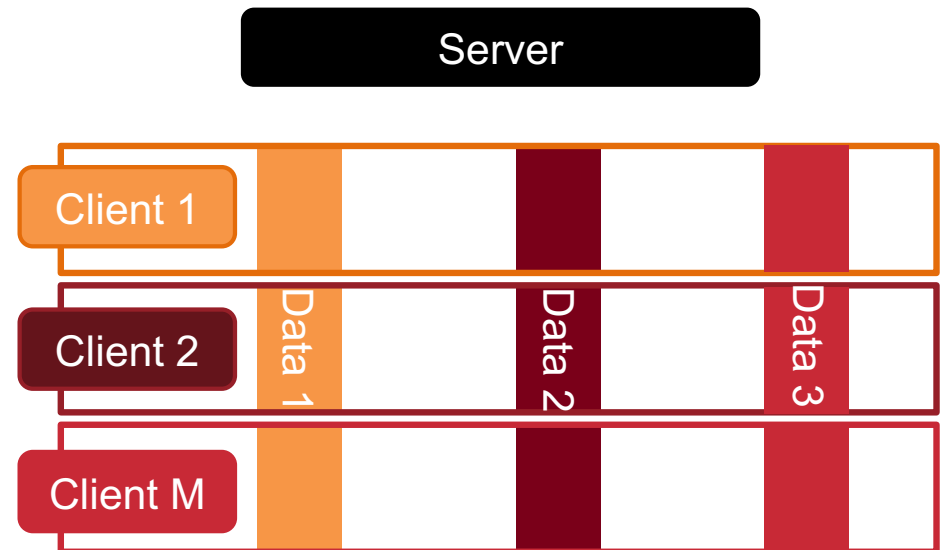
[4] Karimireddy S P, Kale S, Mohri M, et al. Scaffold: Stochastic controlled averaging for federated learning, ICML 2020.



❑ Horizontal & Vertical FL

Vertical FL

- Each client: full data id, **partial** features; **partial** model
- Training done at the server (has access to data)
- Application
 - B2B distributed regression [5]
- Algorithm
 - FedBCD [5], VAFL [6]



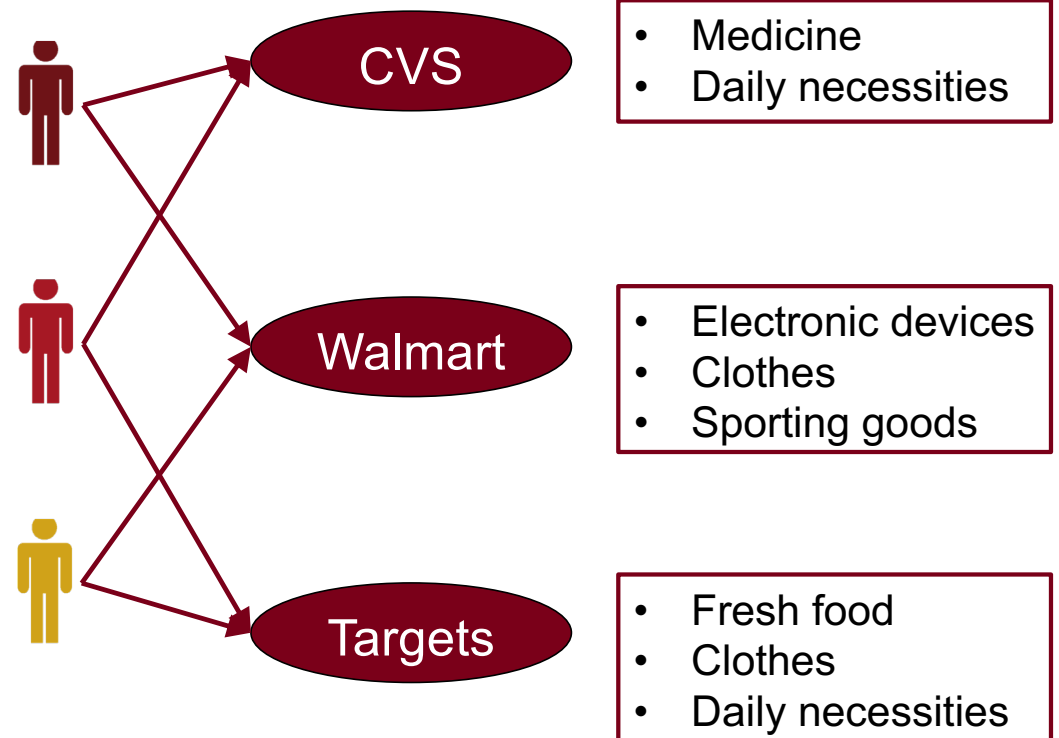
[5] Liu Y, Kang Y, Zhang X, et al. A communication efficient vertical federated learning framework, arXiv 2019.

[6] Chen T, Jin X, Sun Y, et al. VAFL: a Method of Vertical Asynchronous Federated Learning. arXiv 2020.

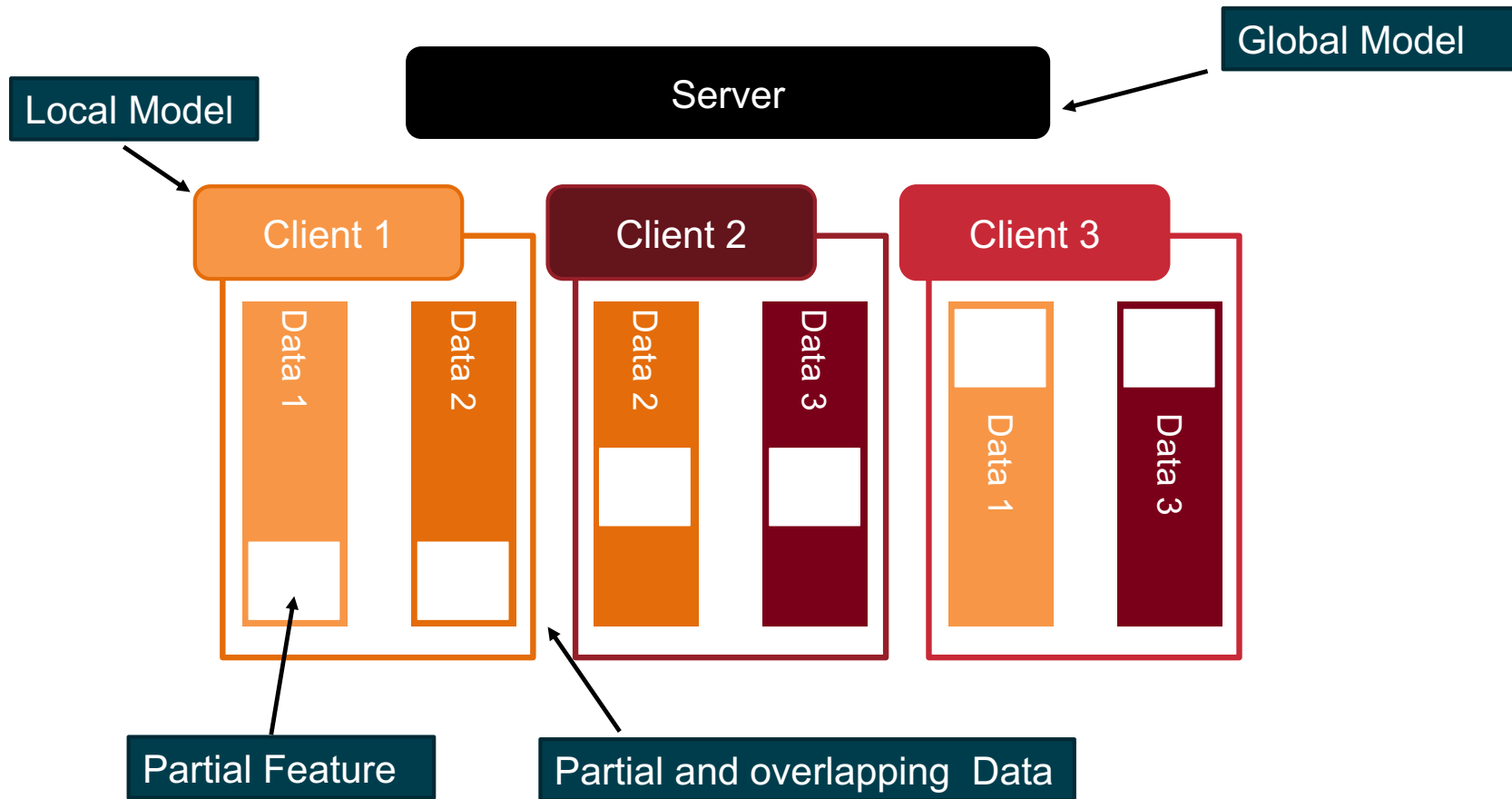


❑ Hybrid FL

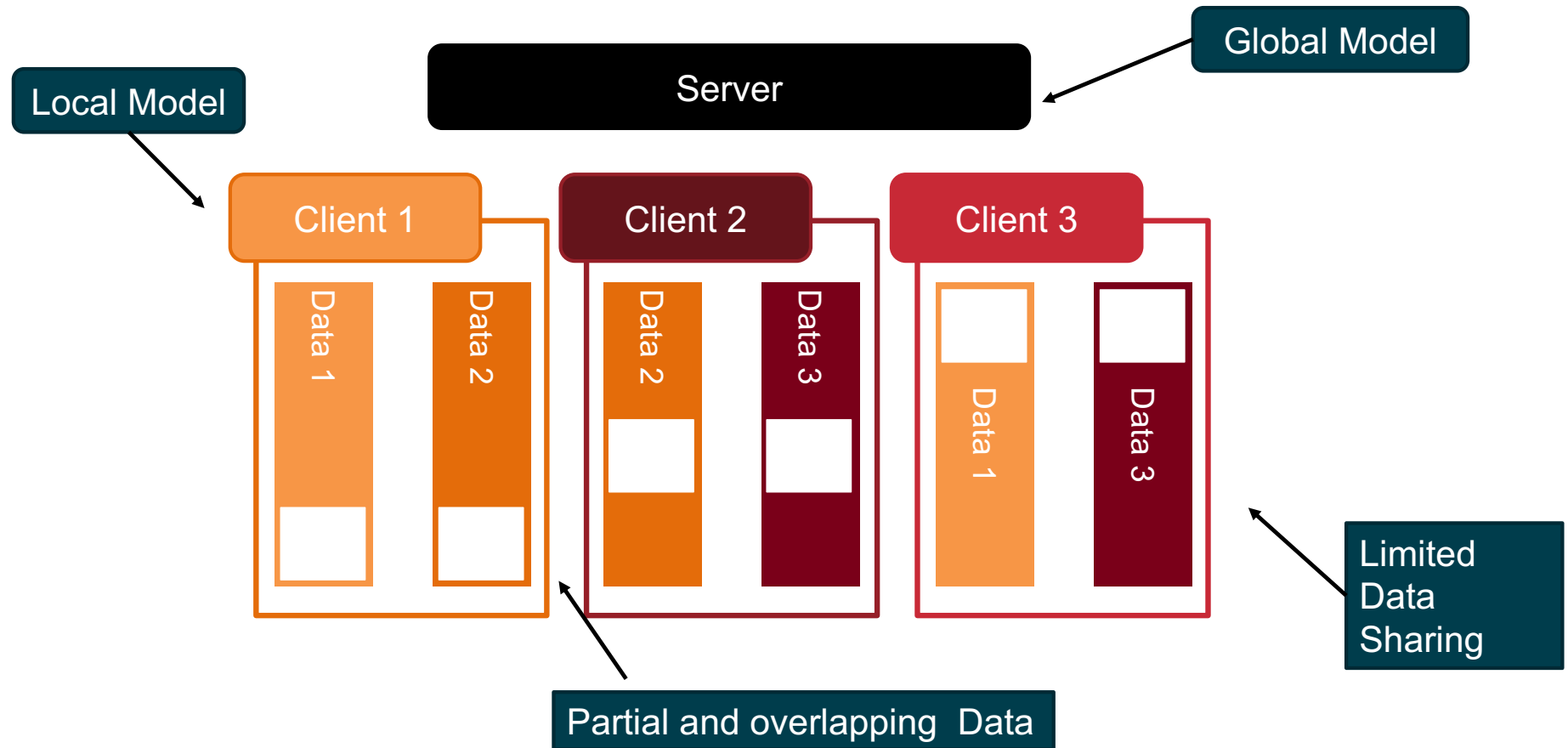
- **Partial** feature & **partial** data id
- Supermarkets:
 - Client: Walmart, Amazon, Ebay, Targets
 - Sample index: costumers
 - Feature: shopping records on different categories of products
 - Goal: jointly predict customers' behavior
- **Not covered by HFL & VFL**



❑ Hybrid FL: Setting



❑ Hybrid FL: Challenges



□ Our Contribution

- Propose a new and flexible Hybrid FL problem formulation.
- Develop an efficient algorithm for Hybrid FL
 1. enables knowledge sharing
 2. maintains data locality
 3. without requiring sample synchronization
- Numerical results on the real data set show competitive performance to centralized training.



□ Hybrid FL Model Design

- System Design
- Problem Formulation
- Regularizer Design

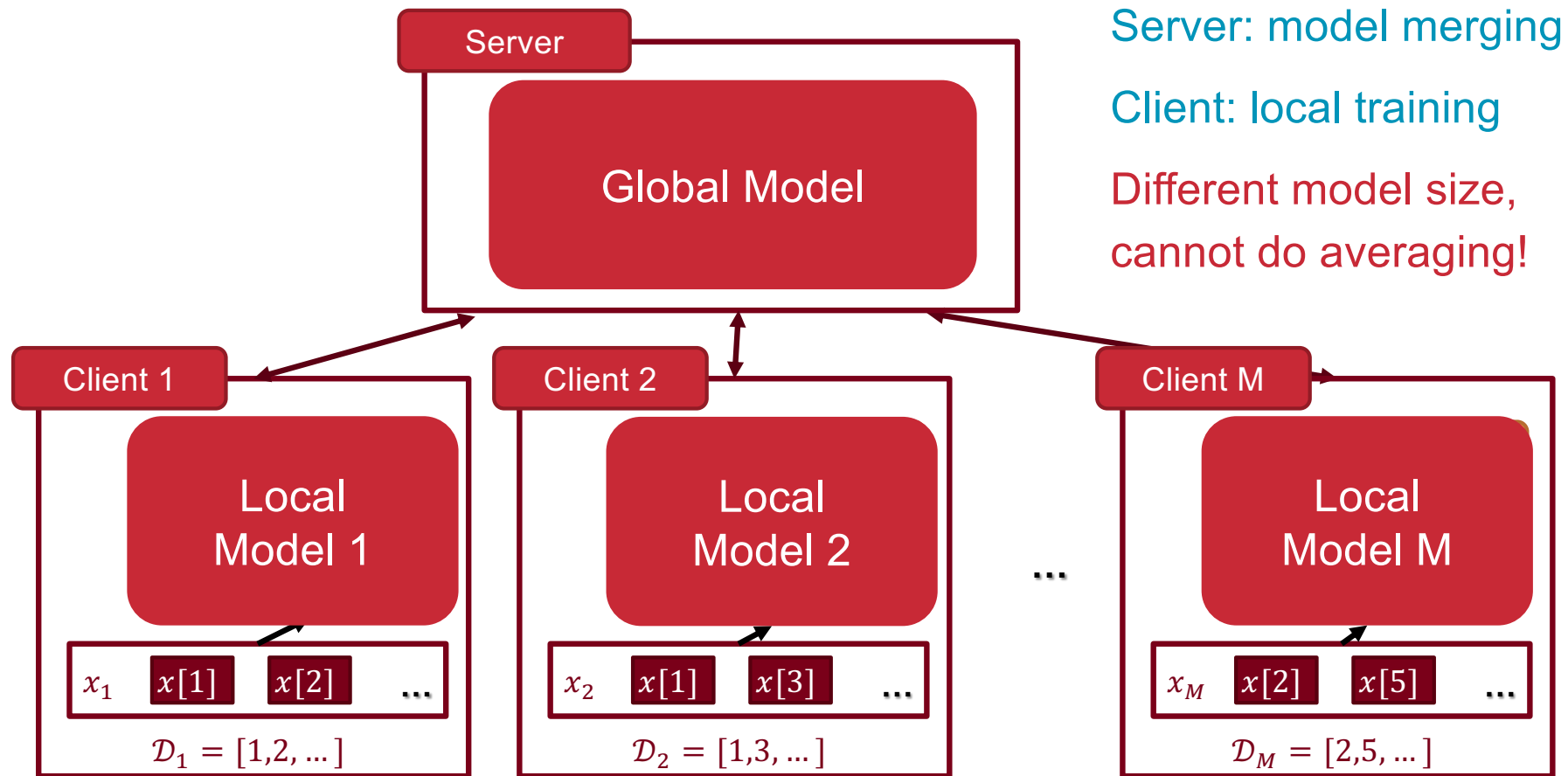


❑ System Structure

- Recall:
 - Challenge 1: no data sharing
 - Challenge 2: build both local and global models
 - Challenge 3: sample synchronization
- Our solution:
 - Local training + Model merging



❑ System Model



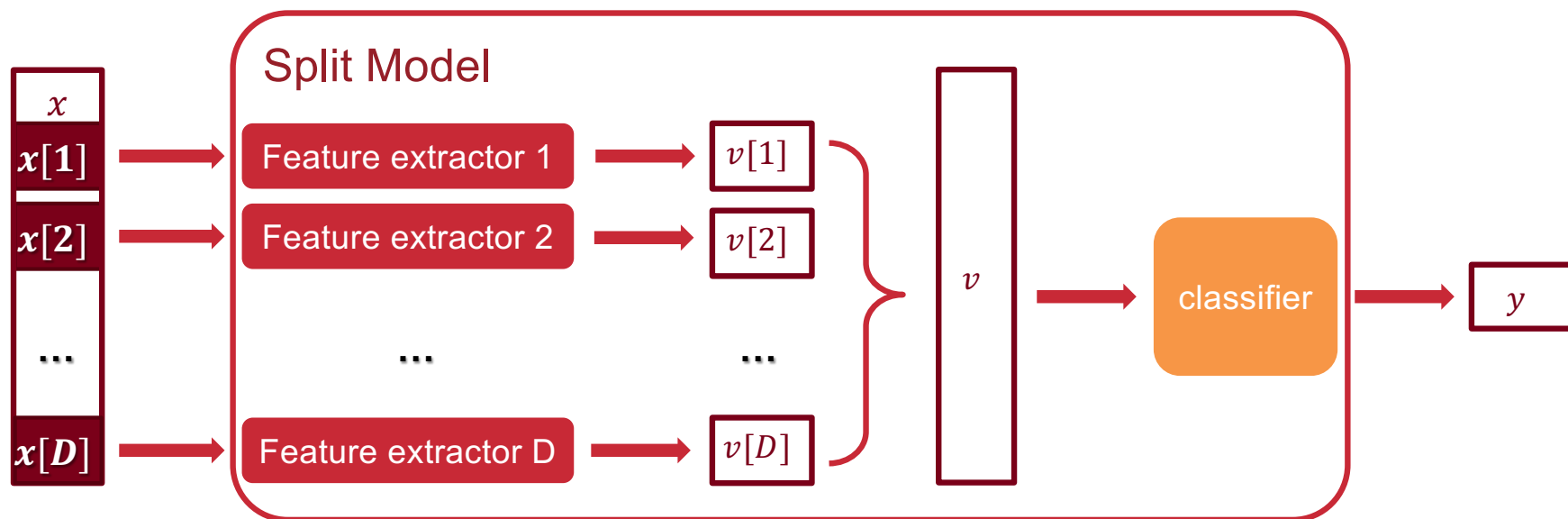
❑ How to Merge Models?

- Global models and local models have different sizes
- However, they have overlapping functionality blocks
- Our Strategy: **Identify** and **Match** the blocks with similar functionalities between client and server

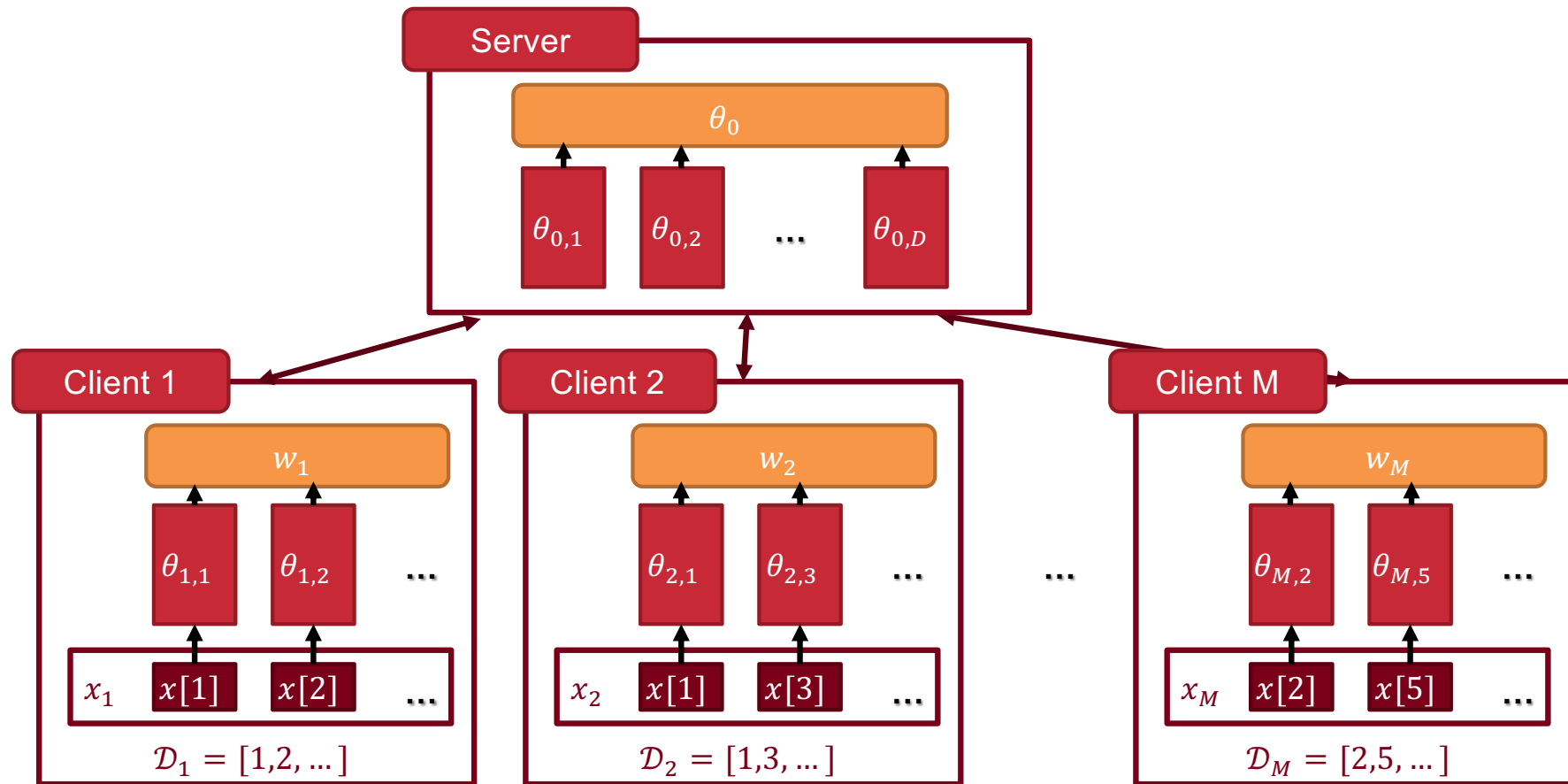


❑ How to Identify Similar Functionalities?

- **Assumption:** Same feature should be processed by the same **feature extractor**

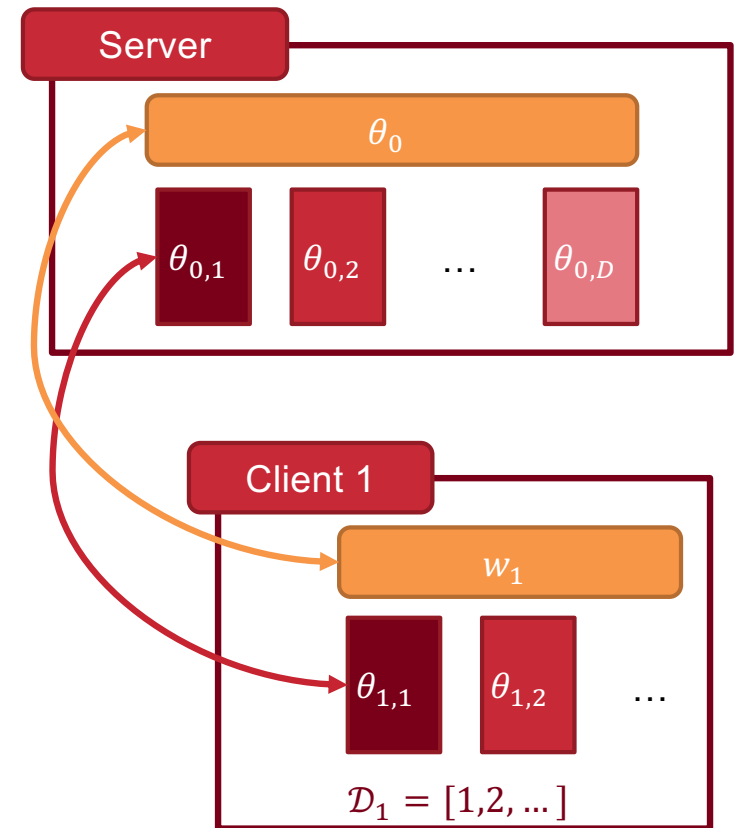
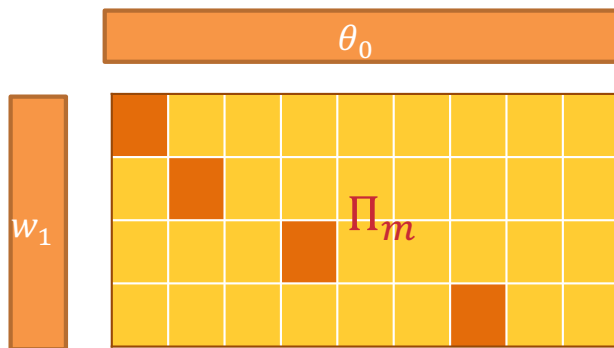


□ A Refined System Model



□ How to Match?

- Extractors of the same feature should be similar: $\theta_{m,d} \approx \theta_{0,d}$, $d = 1, \dots, D$
- The global classifier is an (linear [7]) ensemble of the local ones; needs to find the best matching matrix Π_m



[7] Wang H, Yurochkin M, Sun Y, et al., *Federated Learning with Matched Averaging*, ICLR 2020.

□ Problem Formulation

$$\min_{\{\{\theta_{m,d}\}, w_m\}} \underbrace{\frac{1}{MN} \sum_{m=1}^M \sum_{n \in \mathcal{N}_m} \ell_m(h_m(\{\theta_{m,d}\}; x_m^n); w_m; y^n)}_{\text{Classification loss}} + \underbrace{r(\{\{\theta_{m,d}\}, w_m\}; \theta_0, \{\theta_{0,d}\})}_{\text{Model matching loss}}$$

- Classification part optimizes local models by the local agents;
- Regularizers part matches extractor and classifiers
 - Enable knowledge sharing
- Optimization variables:
 - extractors $\{\theta_{m,d}\}_{d \in \mathcal{D}_m}$
 - matching patterns Π_m
 - classifiers θ_0, w_m



□ Hybrid FL Algorithm Design

- Problem Separation
- HyFEM Algorithm
- Algorithm Analysis



□ Problem at Client

- Local problem at each client m :

$$\min_{\theta_m, w_m} f_m(\theta_m, w_m) = \sum_{n \in \mathcal{N}_m} \ell_m(h(\theta_m; x_m^n); w_m; y_m^n) \\ + \sum_{d \in \mathcal{D}_m} \text{dist}(\theta_{m,d}, \theta_{0,d}) + \mu \text{dist}(\Pi_m \theta_0, w_m);$$

Feature extractor regularizer

Local classification loss

Classifier regularizer

- Use **local data**, **global model** to update local model



□ Problem at Server

- Global problem at server:

$$\begin{aligned} \min_{\theta_0, d} \sum_{m=1}^M \sum_{d \in \mathcal{D}_m} \text{dist}(\theta_{m,d}, \theta_{0,d}); & \quad \leftarrow \text{Feature extractors regularizer} \\ \min_{\theta_0, \{\Pi_m\}} \sum_{m=1}^M \text{dist}(\Pi_m \theta_0, w_m), & \quad \leftarrow \text{Classifier regularizer} \\ \text{s.t. } \Pi_m \mathbf{1} = \mathbf{1}, m = 1, \dots, M. \end{aligned}$$

- **No** local data is used!

❑ Hybrid Federated Matching Algorithm (HyFEM)

Block-type Update

For $t = 0, \dots, T - 1$ do

For each client in parallel does

$$\{\theta_{m,d}^{r,Q}\}, w_m^{r,Q} = \arg \min f_m(\{\theta_{m,d}\}, w_m)$$

Local updates
for Q iterations

For the server does

$$\theta_{0,d}^{t+1} = \frac{1}{|\{m: d \in \mathcal{D}_m\}|} \sum_{m: d \in \mathcal{D}_m} \theta_{m,d}^{t,Q}$$

Merge the feature
extractors

Iteratively do

1. Pick client index m'
2. $\Pi_{m'} = \arg \min_{\Pi_{m'}} \text{dist}(w_m, \Pi_{m'} \theta_0)$
3. $\theta_0 = \arg \min_{\theta_0} \sum_{m=1}^M \text{dist}(w_m, \Pi_m \theta_0)$

Merge the classifiers



❑ Algorithm Analysis

1. Server does **NOT** have direct access local data
2. **Avoids** sample synchronization issue
3. Local models deal with local data with partial features
4. Global model deals with the full featured data

Under mild condition on ℓ_m 's, the algorithm converges by extending the proof of the BCD algorithm [Razaviyayn et al 13].

Razaviyayn M, Hong M, Luo Z Q. A unified convergence analysis of block successive minimization methods for nonsmooth optimization, SIAM Journal on Optimization 2013.



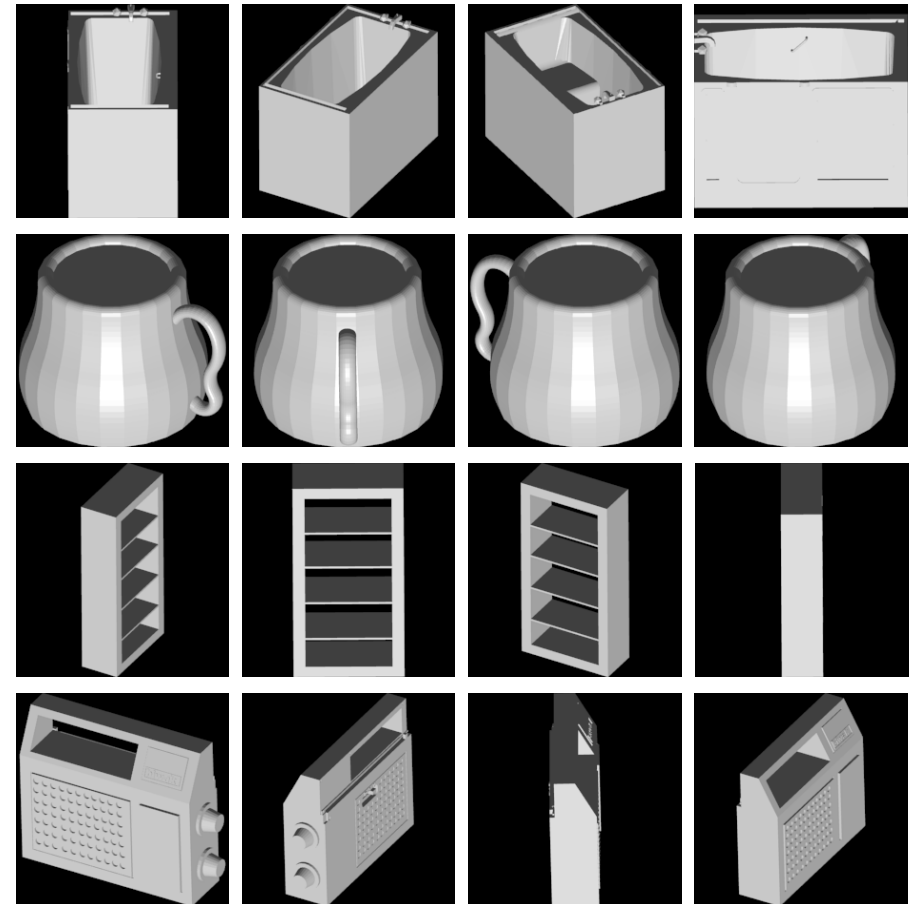
☐ Numerical Results

- Experiment Design
- Parameter Settings
- Results



❑ Experiment Design

- ModelNet40
 - 40,000 train + 4000 test data
 - 40 classes
 - 12 views/sample (feature blocks)
- Model
 - Feature extractor: CNN layers of ResNet-34
 - Classifier: FC net with 1 hidden layer
- Training set
 - Partial views of partial classes / client
- Testing set
 - Local: partial views of all classes
 - Global: full views of all classes

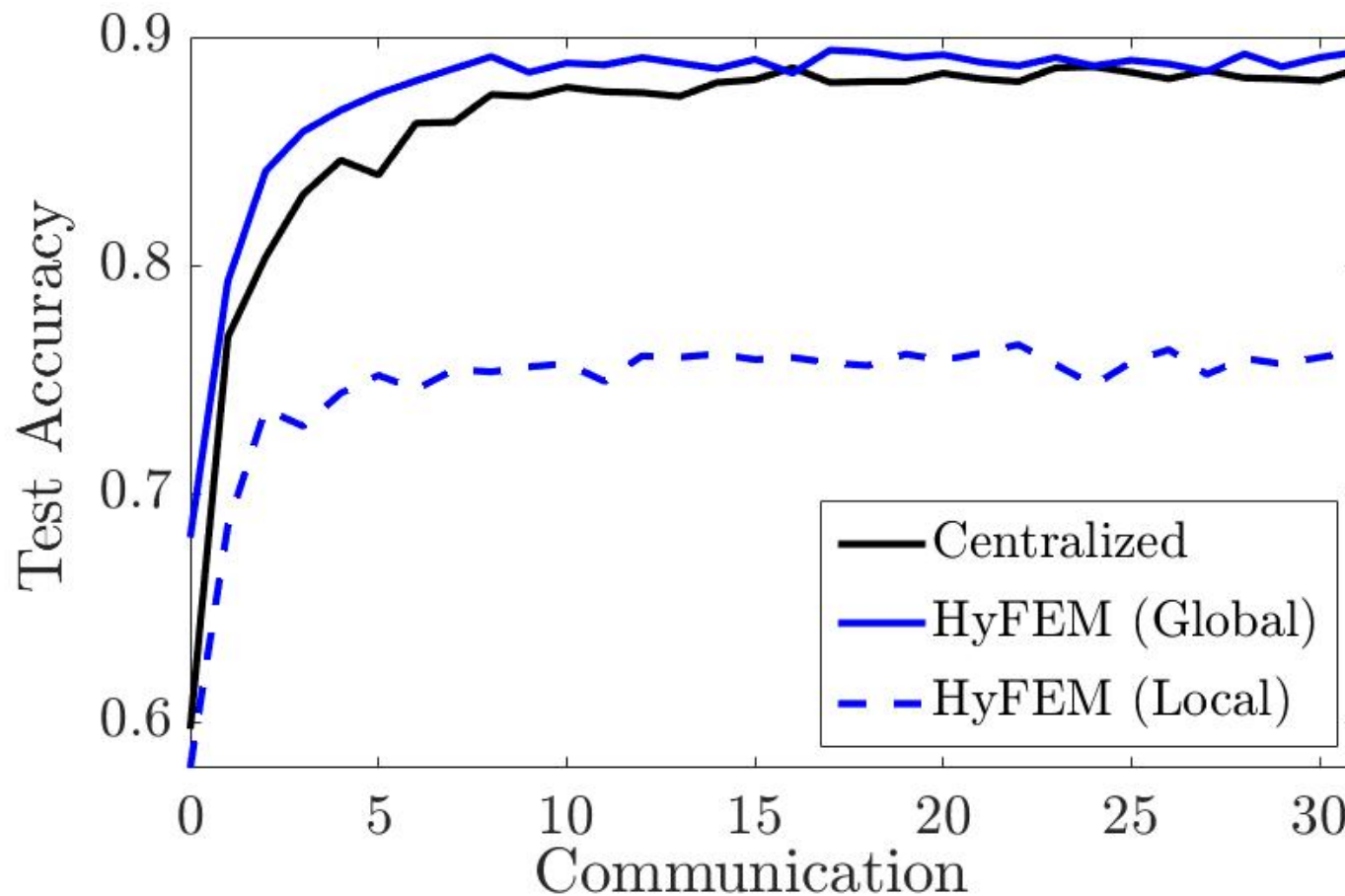


❑ Experiment Settings

Setting	Total view $ D $	# Client M	# Views $ D_m $	# Classes	Used Data
1	4	4	2	20	75%
2			3	30	100%
3	12	8	6	15	88%

Views	1	0	2	2	1	2	0	2	3	2	2	0	3	0	1	1	1	0	2	3	2	2	2	1	1	2	1	1	2	2	2	2	2	1	2	0	1	3	1	2	1
	2	0	1	3	1	1	0	2	3	2	2	0	2	0	2	2	1	0	1	2	2	1	1	1	2	1	2	1	2	2	2	3	3	1	1	1	2	2	1	2	2
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	4	1	2	2	1	2	1	1	1	2	1	1	1	1	3	2	2	1	1	2	1	1	2	0	2	2	2	1	1	1	3	2	4	0	1	2	2	1	1	2	1
	5	1	3	1	1	3	1	1	1	2	1	1	2	1	2	1	2	1	2	3	1	2	3	0	1	3	1	1	1	1	3	1	3	0	2	1	1	2	1	2	0
	6	1	2	2	1	2	1	1	1	2	1	1	1	1	3	2	2	1	1	2	1	1	2	0	2	2	2	1	1	1	3	2	4	0	1	2	2	1	1	2	1
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	8	2	0	2	1	0	1	2	2	1	2	1	2	1	2	1	1	0	2	2	1	1	1	2	1	1	3	1	1	2	3	2	3	2	2	3	2	1	0	1	2
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	12	1	1	1	1	1	0	3	2	3	1	2	1	0	3	3	2	1	0	2	2	1	0	1	3	0	3	1	0	1	4	3	4	1	0	2	2	2	0	1	1
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
		Classes																																							

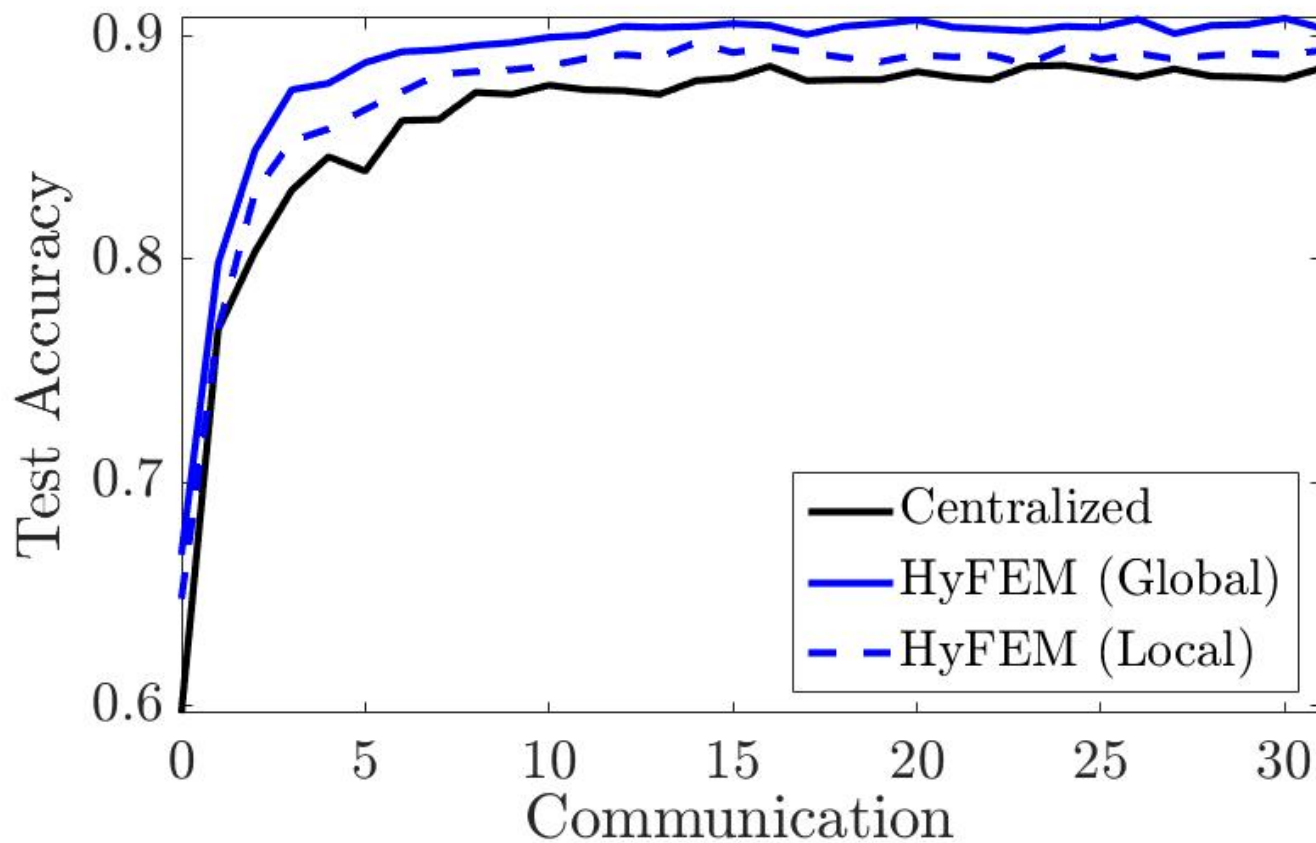
❑ Numerical Result 1 (4 clients, 2 views/client)



Good global model performance

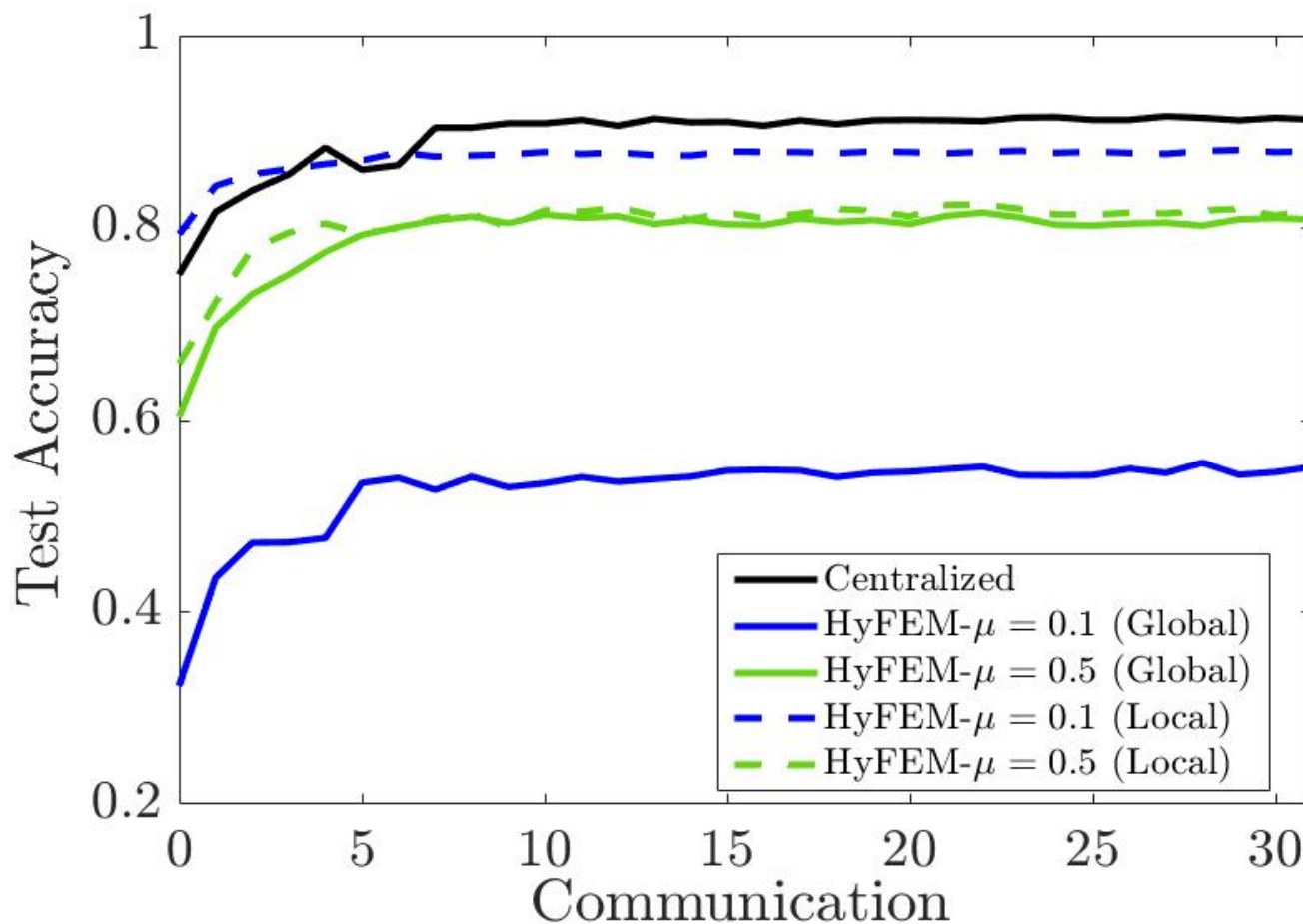
Local feature NOT enough for classification

❑ Numerical Result 2 (4 clients, 3 views/client)



Hybrid FL outperforms centralized training

❑ Numerical Result 3 (8 clients, 6 views/client)



Good local model performance even when client only has half features and 15 classes

Balance between local and global performance by changing μ

❑ Conclusion

1. **Local models** avoid data sharing and sample synchronization.
2. Model **splitting** makes use of the shared features.
3. **Regularizing the classifiers** enables knowledge sharing.
4. Hybrid FL can learn a good global model with incomplete data where HFL and VFL do not apply.
5. Offer an alternative for VFL

